



ÜSKÜDAR UNIVERSITY
FACULTY OF ENGINEERING AND NATURAL SCIENCES
SOFTWARE ENGINEERING DEPARTMENT
Introduction to Brain Computer Interface
2025-2026 SPRING SEMESTER

INSTRUCTOR: Asist. Prof. Dr. Faezeh Rohani

EMAIL: Faezeh.Rohani@uskudar.edu.tr

LECTURE HOURS: Monday: 14:40-17:30

OFFICE HOURS: PLEASE REFER EMAIL FOR APPOINTMENT



Course Description

This course introduces the fundamental concepts of Brain–Computer Interface (BCI) technology.

It covers the main components of BCI systems, different types of BCIs, and their applications in both animals and humans, including medical and non-medical uses.

The course also presents various BCI technologies and emerging trends in the field.

Basic signal processing and machine learning techniques used in BCI development are introduced.

Ethical considerations and future directions, including AI-driven BCIs, are also discussed.



Course Objective

- To introduce the basic concepts and structure of Brain–Computer Interface (BCI) systems.
 - To present different brain signal recording methods and their applications.
 - To introduce basic signal processing and machine learning concepts used in BCI.
 - To discuss applications, challenges, and ethical issues in BCI technology.
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Course Structure and Evaluation

Midterm :

- The proposal submission will count as the midterm evaluation (30%).

Final :

- The final report and presentation together will account for 50% of the final grade.

In-Class Participation (20%):

Active participation in class discussions, weekly exercises.

Project Guidelines

Students must choose **ONE** of the following four project tracks. The objective is to ensure both conceptual understanding and practical exposure to Brain–Computer Interface (BCI) systems.

◆ **Track 1 – Practical Implementation** (Build a system)

- Students design and implement a simple EEG-based BCI system using real data and basic machine learning.
- **Focus:** system design + basic coding + performance evaluation

◆ **Track 2 – Conceptual Design** (Design an idea)

- Students design a conceptual BCI system to solve a real-world problem. No advanced coding is required.
- **Focus:** idea development + system block diagram + scientific reasoning

◆ **Track 3 – Machine Learning Comparison** (Compare models)

- Students compare at least two classifiers using EEG data and analyze their performance.
- **Focus:** understanding model behavior + result analysis

◆ **Track 4 – Research & Publication** Write a research paper

- Students prepare a scientific paper for journal or conference submission.
- **Focus:** academic research + scientific writing

Final Submissions :

All students must submit the following at the end of the semester (according to their selected project track):

- **Final Written Report**

- Clear structure (Introduction, Method/Design, Results or Framework, Discussion, Conclusion)

- **Project Materials**

- Source code (if applicable)
- System block diagram
- Figures, tables, and analysis results

- **Final Presentation**

- presentation
- Clear explanation of the project idea, method, and results



Weekly Schedule

Unit 1 – Foundations of Brain–Computer Interfaces

Week 1 – Introduction to BCI

Definition of BCI

Motivation and historical background

Overview of BCI pipeline

Major application areas

Week 2 – Basic Neuroscience for BCI

Neurons and neural communication

Action potentials

Brain regions and basic functions

Brain activation patterns

Week 3 – Brain Signal Recording & Stimulation

Recording signals from the brain

Invasive vs. non-invasive techniques

Multielectrode arrays

Simultaneous recording and stimulation

Unit 2– Types of BCI Systems

Week 4– Classification of BCI Systems

Invasive BCIs

Non-invasive BCIs

Hybrid BCIs

Advantages and limitations

Week 5– Invasive BCIs

Animal studies

Human implants

Prosthetic arm/hand control

Cursor and robotic control

Week 6 – Non-Invasive- EEG-Based Brain–Computer Interfaces

EEG signal characteristics

ERD/ERS

Event-related potentials (P300, SSVEP – conceptual)

Cognitive paradigms

Practical considerations in EEG-based BCIs

Week 7 fMRI-based BCIs– Other Non-Invasive BCI Modalities

MEG-based BCIs

fNIRS-based BCIs

Hemodynamic vs electrophysiological signals

Temporal vs spatial resolution

Practical comparison with EEG



Weekly Schedule

Unit 3 – Applications, Challenges and Future Directions

Week 8– Medical Applications

Motor restoration

Sensory restoration

Cognitive rehabilitation

Brain-controlled wheelchairs

Week 9– Non-Medical Applications

Gaming

Robotics and avatars

Web navigation

Emotion detection

Week 10 – Neurostimulation and Brain Therapy

Neurofeedback (closed-loop BCI)

TMS

rTMS

tDCS

Closed-loop stimulation systems

Week 11 – Ethics and Future of BCI

Ethical considerations

Implementation challenges

AI-driven BCIs

Wearable systems (Miniaturized and ear-centered EEG technologies)

Future trends

Unit 4 – BCI Signal Processing and Machine Learning

Week 12 – BCI System Structure and Signal Collection

- Signal collection (EEG focus)
- Open-loop vs. closed-loop systems
- system-level feedback in BCI architecture
- Introduction to preprocessing concepts

Week 13 – Brain Signal Processing Fundamentals

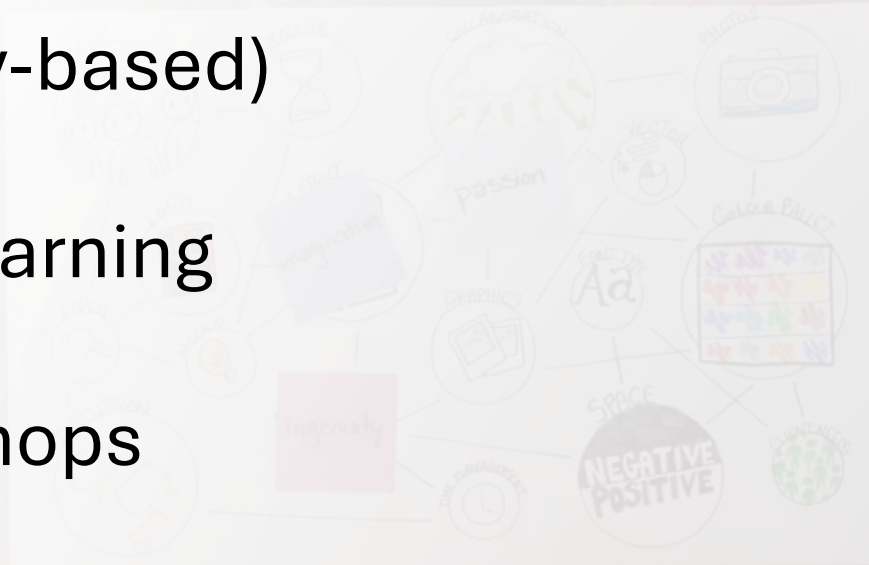
- Preprocessing (artifact removal, filtering)
- Time-domain analysis
- Frequency-domain analysis
- Spectral representation (conceptual)
- Practical EEG processing examples

Week 14 – Feature Engineering and Machine Learning for BCI

- Feature extraction methods (time & frequency features)
- Pattern representation
- Types of classifiers (LDA, SVM, basic neural networks)
- Training vs. testing
- Cross-validation
- Performance evaluation metrics (accuracy, precision, recall)

Teaching Methods

- Lectures (Theory-based)
- Project-based learning
- Practical Workshops
- Group discussions ,Teamwork & peer review sessions
- Final project presentation



Learning Outcomes

By the end of this course, you will be able to:

- Explain the basic concepts and structure of BCI systems.
- Compare different brain signal recording methods.
- Design a simple conceptual BCI system.
- Perform basic EEG signal preprocessing and analysis.
- Apply a basic classification method to brain signal data and evaluate its performance.
- Discuss applications and ethical considerations of BCI systems.

COURSE MATERIALS:

1. Ella Hassianien, A & Azar.A.T, Brain-Computer Interfaces Current Trends and Applications, Springer, 2015, ISBN 978-3319109787
2. Rajesh.P, N.Rao, Brain-Computer Interfacing: An Introduction, Cambridge University Press, First edition, 2013, 978-0521769419
3. Jonathan Wolpaw, Elizabeth Winter Wolpaw, Brain Computer Interfaces Principles and practice, Oxford University Press, USA, Edition 1, January 2012, 978-0199921485
4. Bernhard Graimann, Brendan Allison, Gert Pfurtscheller, Brain-Computer Interfaces: Revolutionizing Human-Computer Interaction, Springer, 2010, 978-3642020902
5. Pachori, Ram Bilas. Time-frequency analysis techniques and their applications. CRC Press, 2023, 9781032392974.
6. Wolpaw, J. R. (2013). Brain-computer interfaces. In Handbook of clinical neurology (Vol. 110, pp. 67- 74). Elsevier.
7. Nam, C. S., Nijholt, A., & Lotte, F. (Eds.). (2018). Brain-computer interfaces handbook: technological and theoretical advances. CRC Press.
8. Wolpaw, J. R., Millan, J. D. R., & Ramsey, N. F. (2020). Brain-computer interfaces: Definitions and principles. Handbook of clinical neurology, 168, 15-23.
9. Clerc, M., Bougrain, L., & Lotte, F. (Eds.). (2016). Brain-computer interfaces 2: technology and applications. John Wiley & Sons.



ÜSKÜDAR UNIVERSITY
FACULTY OF ENGINEERING AND NATURAL SCIENCES
SOFTWARE ENGINEERING DEPARTMENT
Large Language Models
2025-2026 SPRING SEMESTER

INSTRUCTOR: Asist. Prof. Dr. Faezeh Rohani

EMAIL: Faezeh.Rohani@uskudar.edu.tr

LECTURE HOURS: Wednesday: 8:40-11:30

OFFICE HOURS: PLEASE REFER EMAIL FOR APPOINTMENT

Course Description

- This course introduces Large Language Models (LLMs) in a clear and structured way. It covers the basic theory behind LLMs, how they are trained, and the latest developments in the field.
- Topics include transformer structure model, pre-training and instruction tuning, scaling, retrieval-augmented generation, efficient fine-tuning methods, multimodal models, evaluation techniques, and ethical issues.
- The course focuses on building strong conceptual understanding, understanding how LLM systems work, and critically analyzing modern LLMs used in research and industry.

Course Objective

- This course aims to provide a structured and system-level understanding of Large Language Models (LLMs), training paradigms, scaling behavior, and real-world deployment considerations.
- Students will develop the ability to critically analyze modern LLM systems, understand alignment and efficiency challenges, and design research-oriented or application-driven solutions using state-of-the-art large-scale models.

Project Guidelines

Students may choose one of the following options:

Option 1 – Research Project

Conduct a structured research study on an LLM-related topic (literature review, comparative analysis, alignment studies, failure analysis, etc.).

Option 2 – Implementation Project

Design and implement a small LLM-based system (e.g., a simple agent, RAG application, evaluation framework).

Option 3 – Course Topic Presentation

Select one chapter or major topic from the course syllabus and deliver a full lecture-style presentation.

The presentation session may be conducted online if needed.

Weekly Schedule

Week 1 – Introduction to Large Language Models

- Evolution from NLP to LLMs
- Why LLMs are fundamentally different
- Capabilities and limitations

Week 2 – Language Modeling Foundations (Review)

- Language modeling objectives
- From statistical to neural language models (conceptual overview)
- Motivation for large-scale models

Week 3 – Neural Language Models and Sequence Modeling

- RNNs, LSTMs, Seq2Seq models (high-level review)
- Limitations of recurrent architecture

Week 4 – Transformer structure

- Self-attention and multi-headed attention
- Positional encoding
- Encoder-only, decoder-only, and encoder–decoder architectures

Week 5 – Training Large Language Models

- Pre-training objectives
- Data and compute considerations
- Decoder-only models (e.g., GPT-style)

Week 6 – Emergent Abilities and In-Context Learning

- Emergence in large-scale models
- Few-shot and zero-shot learning
- Model-centric explanation of in-context learning

Week 7 – Instruction Tuning and Alignment

- Instruction tuning paradigms
- Reinforcement Learning from Human Feedback (RLHF)
- Alignment challenges and limitations

Week 8 – Scaling Laws and Model Efficiency

- Scaling laws in LLMs
- Compute–data–performance trade-offs
- Limits of scaling



Weekly Schedule

Week 9 – Retrieval-Augmented Language Models

- Motivation for retrieval
- RAG architecture
- Examples: REALM, RAG, FiD (conceptual overview)

Week 10 – Parameter-Efficient Adaptation

- Fine-tuning vs parameter-efficient methods
- LoRA and prefix tuning (conceptual)
- Practical considerations

Week 11 – Tool-Augmented LLMs and Agents

- LLMs with external tools
- Function calling and agentic workflows
- System-level design considerations

Week 12 – Multimodal Large Language Models

- Vision–language models
- Multimodal pre-training strategies
- Applications and challenges

Week 13 – Evaluation, Safety, and Ethics

- Evaluation of benchmarks and challenges
- Hallucination and robustness
- Bias, toxicity, and responsible AI

Week 14 – Recent Advances and Student Projects

- Overview of recent models (GPT, LLaMA, Claude, Gemini)
- Student project presentations
- Course wrap-up

Course Structure and Evaluation

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In-Class Participation (20%):

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Final Submissions :

All students must submit the following at the end of the semester (according to their selected project subject):

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- Clear structure (Introduction, Method/Design, Results or Framework, Discussion, Conclusion)

- **Project Materials**

- Source code (if applicable)
- System block diagram
- Figures, tables, and analysis results

- **Final Presentation**

- presentation
- Clear explanation of the project idea, method, and results

Learning Outcomes

By the end of this course, students will be able to:

1. Explain the evolution from traditional NLP systems to modern Large Language Models.
2. Describe the transformer structure and its role in LLMs.
3. Understand pre-training, instruction tuning, and alignment strategies used in LLM development.
4. Analyze emergent capabilities such as in-context learning.
5. Explain scaling laws and efficiency trade-offs in large-scale models.
6. Understand retrieval-augmented and tool-augmented LLM systems.
7. Critically evaluate modern LLMs in terms of performance, safety, and ethical impact.
8. Conduct analytical or experimental projects related to LLM architecture and applications.

Teaching Methods

- Lectures (Theory-based)
- Project-based learning
- Practical Workshops
- Group discussions ,Teamwork & peer review sessions
- Final project presentation

COURSE MATERIALS:

- Chakraborty, T. (2025). *Introduction to Large Language Models*. Wiley.
- Jurafsky, D., & Martin, J. H. *Speech and Language Processing*.
- Eisenstein, J. *Introduction to Natural Language Processing*.
- Selected papers from ACL, EMNLP, NAACL, NeurIPS, ICML, ICLR, AAAI, TACL, JMLR.